D3PM: Dual-Stream Denoising Diffusion Probabilistic Model for Change Detection in Multimodal Remote Sensing Images

Fenlong Jiang, *Member, IEEE*, Xinlong Huo, Mingyang Zhang, *Member, IEEE*, Maoguo Gong, *Fellow, IEEE*, Yan Pu, *Graduate Student Member, IEEE*, Yu Zhou, Wei Zhao, *Member, IEEE*, and Ziyu Guan, *Member, IEEE*

***Abstract*— Detecting land cover changes from multitemporal and multimodal remote sensing images acquired by different sensors at the same location is a complex yet highly valuable task. Recently, diffusion models, exemplified by the denoising diffusion probabilistic model (DDPM), have garnered significant attention for their remarkable performance and straightforward architecture. These models excel in image generation, distribution modeling, and feature extraction, making them highly promising for advancing multimodal change detection (MCD). In this article, we propose a dual-stream DDPM (D3PM) to address the challenges of MCD. Specifically, D3PM leverages DDPM to design two distinct processing streams, one for each image modality. The first stream employs an unconditional DDPM, whose denoising encoder–decoder network can achieve robust feature extraction. The second stream employs a conditional DDPM to facilitate modal translation, enabling the extracted features to align with the characteristics of the other modality, thereby improving cross-modality comparability. To further enhance performance, we constructed a CD task branch based on the decoder features of the two DDPMs across multiple denoising time steps. In addition, we designed a collaborative learning optimization strategy with asynchronous time steps, fostering cross-task knowledge sharing and mutual enhancement while preserving the integrity of individual task learning. Experimental results on multiple public datasets demonstrate the effectiveness and superiority of the proposed D3PM, which achieves efficient modal transformation and alignment, mitigates modal heterogeneity interference, and significantly improves detection performance.**

***Index Terms*— Change detection (CD), denoising diffusion probabilistic model (DDPM), multimodal remote sensing images, multitask learning.**

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Fenlong Jiang, Wei Zhao, and Ziyu Guan are with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, School of Computer Science and Technology, Xidian University, Xi'an 710071, China (e-mail: jiangfenlong@outlook.com; ywzhao@mail.xidian.edu.cn; zyguan@xidian.edu.cn).

Xinlong Huo is with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China (e-mail: xinlong@stu.xidian.edu.cn).

Mingyang Zhang, Yan Pu, and Yu Zhou are with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: omegazhangmy@gmail.com; pu7yan9@gmail.com; zhouyu@xidian.edu.cn).

Maoguo Gong is with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, School of Electronic Engineering, Xidian University, Xi'an 710071, China, and also with the Academy of Artificial Intelligence, College of Mathematics Science, Inner Mongolia Normal University, Hohhot 010011, China (e-mail: gong@ieee.org).

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I. Introduction

CHANGE detection (CD) is a challenging remote sensing task that predicts the change in land cover within a region by comparing two images of the same geographic area captured at different times [1], [2]. Change maps obtained by CD can accurately reflect changes of the actual surface, and these images play a vital role in monitoring and understanding changes on the Earth's surface. CD techniques have an extensive range of applications including: flood extraction [3], climate environmental monitoring [4], land cover management [5], and urban change analysis [6].

The comparability between remote sensing data plays a crucial role in obtaining accurate and timely CD maps from bitemporal images. In this article, some interfering factors that prevent direct comparison of the images, such as the noise generated during imaging [7], spatial and temporal image resolution disparties [8], radiometric and atmospheric variations [9], and influence of image registration [10], may reduce the accuracy of CD. In order to be affected by these factors as little as possible, scholars have proposed many methods using single-modal data for CD, which are usually based on the assumption that these data have homogeneity and high comparability. Traditional methods usually obtain change maps by directly analyzing the spectral differences of multitemporal images, such as the classic change vector analysis [11] and multivariate CD methods [12]. With the development of artificial intelligence, scholars have explored CD methods based on deep learning, which usually regard CD as an end-to-end multiclassification problem [13], [14]. Although many single-modal CD methods have achieved success, however, in the face of multisource and multitarget scenarios in real life, it is difficult to determine which kind of remote sensing image data is more conducive to balancing the effectiveness and accuracy of CD results [15]. In fact,

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multimodal images bring richer features, such as speckle noise in synthetic aperture radar (SAR) images, multiple bands in multispectral images, and mixed pixels in hyperspectral images.

The development of spatial satellites equipped with various sensors has brought both a rich data source and more challenging issues for the remote sensing and computer vision community [16]. Multimodal images provide more information about sensors, angles, resolution and time compared to singlemodal, which provide more observation details to refine the scene information and bring a huge boost to the cutting edge. At the same time, since heterogeneous images from multiple sensor types undermine the assumption of high comparability, multimodal data also pose high requirements for the effectiveness of change information extraction. To address the above issues, a variety of multimodal CD (MCD) methods that can be broadly classified into two major categories have been proposed. An approach follows the idea of regarding CD as an end-to-end multiclassification task and usually directly deals with multimodal data, aligns their mappings in the high-dimensional space and finally uses the aligned features to predict CD results [17], [18]. As the data alignment is conducted at the abstract feature level, the interpretability of these methods is relatively limited. Moreover, the network models employed in these methods tend to be more intricate and larger in scale compared with those used in single-modal CD. Another intuitive method is to implement a consistent observation domain through image translation. For example, using generative adversarial network (GAN)-based networks to translate one type of image into another type, modifying heterogeneous images into isomorphic ones, so that multimodal data can be regarded as singlemodal data, and then, the CD results can be obtained directly in their common observation space [19], [20], [21]. However, these two-stage methods did not consider how to extract suitable image representations by treating the processing of multimodal data and CD as a whole work. In addition, they focus too much on pixel-level translation to obtain visually unified images, which usually leads to inaccurate feature mapping between multimodal data and the loss of important semantic information [22], and then cause cumulative errors.

In recent years, diffusion models have increasingly emerged as a trending topic in the field of computer vision. Diffusion models are generative models, and their training is usually based on two stages: a forward diffusion stage in which Gaussian noise is gradually added to transform the input data into pure noise; and a reverse diffusion stage in which data samples are restored from pure noise. Meanwhile, in the reverse stage, whether additional conditional information is introduced to guide the denoising process determines whether the model is conditional or unconditional in generation. Diffusion models excel in terms of generation quality and stability, primarily due to their ability to operate directly in the data space without the need to explicitly construct a latent space. By progressively denoising, the model can more freely learn the data distribution, thereby avoiding the distortions caused by constraints in the latent space, which are commonly seen in generative models such as GANs and variational autoencoders (VAEs). However, they still have room for improvement in computational efficiency and real-time performance. In comparison, GANs are more advantageous in terms of generation speed and efficiency, but their training processes tend to be unstable; VAEs, while stable in training, often produce more blurry samples. Despite these limitations, diffusion models still demonstrate state-of-the-art achievements in many fields, particularly in conditional generation [23], [24] and image-to-image translation [25], [26].

More importantly, some research has constructed denoising pretraining tasks for unconditional diffusion models, allowing the network to perform unsupervised learning on a large amount of unlabeled data. After that, the network is fine-tuned on the labeled data, in order to accomplish various computer vision tasks [27], [28], [29]. This proves that the unique training mode of diffusion models can endow networks with powerful feature extraction capabilities. Inspired by above works, many scholars have explored the application of unconditional diffusion models in singlemodal CD and obtained excellent results [30], [31]. Concurrent works have also explored the idea of conditional generation, designing conditions to guide the generation of CD maps [32], [33]. However, in the field of MCD, the potential of both conditional and unconditional diffusion models has not been fully and promptly exploited.

In this article, we deploy the denoising diffusion probabilistic model (DDPM) [34] in MCD tasks and simultaneously combine conditional and unconditional DDPMs to design a multitask MCD framework named dual-stream DDPM (D3PM) MCD, with the aim of mitigating the impact of data heterogeneity on CD and further exploring the full utilization of multimodal data. It not only can reduce modal heterogeneity and align data features but also can avoid the cumulative errors and time costs of multistage methods. Our focus lies in leveraging the forward process of the DDPM to construct a multitask framework for image denoising and translation, and a collaborative learning optimization strategy with asynchronous time steps is designed for this framework. Through multitask collaborative feature extraction, it facilitates the transformation and alignment of cross-modal features. Ultimately, high-precision CD is achieved by integrating features across multiple time steps and multiple scales. Specifically, D3PM makes use of DDPM to create a dual-stream branch encoder for two image modalities. Using optical and SAR image pairs as examples, the image-denoising U-Net stream (IDUS) employs an unconditional DDPM, whose network can achieve robust feature extraction by denoising the optical image. The image-translating U-Net stream (ITUS), based on a conditional DDPM, extracts high comparable across modalities features by translating the SAR image to the optical image. In addition, we deployed a lightweight CD head to predict the results using features of the two DDPMs. Our experimental results demonstrate that when the dual-stream branch DDPM encoder is applied for CD, it performs better than the state-of-the-art CD methods. This further illustrates that the image-to-image translation and feature extraction capabilities of DDPM can be applied to produce useful representations for CD.

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In summary, the contributions of D3PM can be summarized as follows.

1) We propose a D3PM, where one stream is an unconditional DDPM and the other is a conditional DDPM. This approach enables robust feature extraction of multimodal data and endows the features with a high level of cross-modality comparability.
2) We designed a feature fusion mechanism with multiple denoising time steps for the CD task, aiming to fully utilize multiscale and multilevel features to obtain high-precision prediction results.
3) A multitask collaborative learning strategy with asynchronous time steps was designed. It promoted data and knowledge sharing across tasks.
4) Extensive experimental results on MCD datasets demonstrated that the proposed method outperforms existing state-of-the-art CD methods in terms of CD effectiveness.

The rest of this article is organized as follows. Section II overviews existing singlemodal, multimodal, and diffusion model-based CD methods. In Section III, we delve into a detailed description of the proposed method. Section IV describes our experimental setup, compares the results obtained by proposed method with existing methods both qualitatively and quantitatively and conduct ablation experiments. Finally, in Section V, we presents our conclusions and outlook for future work.

II. Related Works

A. Single-Modal CD

Single-modal images are the most commonly used data sources for CD. Some early works targeting small-resolution images usually took pixels or pixel neighborhoods as units and classified them to generate change maps. Huang et al. [35] used the support vector machine (SVM) model to detect changes in vegetation coverage. Mastro et al. [36] employed the random forest model to extract the areas affected by floods in SAR images before and after disasters. Benedek et al. [37] designed a multilayer Markov random field model for CD in optical remote sensing images. However, these methods have poor generalization ability and are difficult to apply to large-scale datasets with high resolution.

The tremendous success of deep learning in the field of computer vision has promoted the development of CD methods based on convolutional neural networks (CNNs). Most of these methods are end-to-end; that is, they directly obtain change maps by processing bitemporal images. The pioneering work was the three models proposed by Daudt et al. [38], which are input image fusion, feature concatenation fusion, and feature difference fusion, respectively. Later, many methods introduced the attention mechanism to enhance the task-related features and suppress the task-unrelated ones. Noman et al. [39] captured the enhanced global context and local spatial information through pooling-transpose (PT) attention and deep convolution, and then fused the two to obtain robust features; Fang et al. [40] designed an attention module that can extract the most representative features at different semantic levels and reduced the loss of deep information in neural networks through dense connections; Chen and Shi [13] designed a spatial–temporal attention neural network based on the feature pyramid to capture the spatial–temporal dependencies of ground objects of different sizes.

Transformer [41] is a fully connected network that uses self-attention and cross-attention mechanisms. In recent years, it has been widely applied in the area of computer vision and has achieved remarkable success in corresponding tasks. In the field of CD, Chen et al. [42] made an early attempt. They used CNN to extract the initial features of bitemporal images and then used Transformer to conduct context modeling on the spatial and temporal of the initial features to obtain self-attention features. Bandara and Patel [43] designed a Siamese network that uses an Transformer-based encoder and connected a lightweight multilayer perceptron as a decoder to obtain change maps effectively and efficiently. Noman et al. [44] improved the performance of Transformer trained from scratch by utilizing the random sparse attention operation and introduced a change feature enhancement module to enhance the relevant semantic changes while suppress the noisy ones. Inspired by the success of Swin Transformer [45], Zhang et al. [46] designed a encoder and decoder with Swin Transformer blocks as the basic units and achieved excellent performance.

All in all, both CNN-based models and Transformer-based models perform well on single-modal data. However, single-modal CD methods usually deal with highly comparable homologous data, and tend to be limited when applied to the heterogeneous multimodal data.

B. Multimodal CD

There are two predominant approaches in mainstream MCD methodologies: the first category involves aligning multimodal data mappings in high-dimensional spaces and subsequently utilizing these aligned features to generate CD maps, and the second approach employs two-stage frameworks that initially transform multimodal data into visually consistent representations to reduce heterogeneity, followed by CD implementation in the subsequent phase.

Many methods use the first strategy for CD, which can be traced back to some earlier classical works. Mubea and Menz [47] explored the potential benefits of combining data from optical sensors with Radar sensor data from Advanced Land Observing Satellite Phased Array Type L-band SAR data, and applied maximum likelihood (ML) and SVM are applied. Qin et al. [48] stacked two images acquired by different sensors together and principal component analysis (PCA) was applied to the stacked data. Current research has been significantly reshaped by deep learning due to its powerful feature extraction ability, Hafner et al. [49] proposed a U-Net-based dual-stream architecture for optical–SAR CD. The network extracts deep features using two separate U-Nets, then combines multimodal features by simple concatenation. He et al. [50] design different structured independent encoders for Sentinel-2 and Sentinel-1 images to extract features, and performs features fusion at multiple stages using gating

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and self-attention modules. Furthermore, they overcome the feature misalignment issue during decoding by embedding a feature alignment module in the upsampling operation. Zhang et al. [17] proposes a domain adaptation-based multisource CD network to process heterogeneous optical and SAR images, and employs feature-level translation to align inconsistent deep feature spaces in heterogeneous data. Benefiting from this design, feature space translation and CD are bridged within the network to encourage task communication. Pu et al. [51] proposed an adversarial feature equilibrium network that establishes an additional adversarial optimization to solve the equilibrium problem between modal differences and land–cover changes while effectively integrating change information. Han et al. [52] extracted dual-temporal similarity feature maps with high similarity by learning the joint features of heterogeneous images, then mapped the feature maps to an approximate feature space for comparison through joint feature learning optimized by similarity measurement. Finally, the change map was obtained by calculating the differences of the feature maps. Han et al. [53] also proposed a global structure graph mapping method that constructs a global structure graph to represent the structural information of multimodal images and then cross-maps it to another data domain to obtain "comparable" structural features. However, the aforementioned methods for addressing data heterogeneity demonstrate limited interpretability, as they remain challenging to elucidate the underlying mechanisms enabling feature alignment through algorithmic design.

Inspired by the success of image translation in the field of natural images [54], [55], some studies focused on homogenization methods for heterogeneous images, which consist of the translation of images with different distribution characteristics into the same latent feature space, or enable multimodal images to achieve visual uniformity. Niu et al. [20] adopted a conditional GAN (cGAN) to translate the heterogeneous SAR and optical images into some space, where their information has a more consistent representation, making the two translated and approximated images can be considered as homogeneous, from which the final change map can be acquired by direct comparison. Liu et al. [56] proposed heterogeneous remote sensing CD by pixel translation and then fuzzy clustering to detect the changed regions with pixel-level difference discrimination. Saha et al. [21] used a cycle-consistent GAN (CycleGAN [57]) to translate SAR images into optical images. After the network is trained, the generator is used as a feature extractor to extract optical-like features from bitemporal SAR images, and then apply deep change vector analysis (DCVA) and fuzzy rules to identify changed regions. In order to mitigate the impact of the intertwining of modal heterogeneity and changes in ground objects during the processing on CD, Jiang et al. [58] proposed a change-masked modality alignment network. It employs two Image modal transformation branches to conduct conversions between data modalities and utilizes the change probability map predicted by the CD branch to mask the changed areas in Image Modal Transformation branches, further decoupling the changes in ground objects from modal heterogeneity. While these methodologies appear conceptually straightforward, excessive emphasis on achieving visually coherent transformation outputs often results in imprecise feature mapping and compromised semantic information integrity, ultimately leading to error propagation throughout the analytical pipeline.

Although both paradigms have demonstrated measurable success, fundamental scientific challenges persist in MCD research: specifically, how to effectively minimize heterogeneity-induced interference while simultaneously achieving feature alignment within an integrated framework that concurrently addresses CD and multimodal data transformation.

C. Diffusion Model-Based CD

Diffusion models have potential that cannot be underestimated in the field of remote sensing image CD. CD methods based on diffusion models are usually divided into two major categories: one focuses on the efficient training strategies of unconditional diffusion models and uses the trained network as a feature extractor; the other emphasizes the powerful generation ability of conditional diffusion models, taking bitemporal images as conditions to guide the denoising process and directly generate CD maps.

Some work in the field of remote sensing has demonstrated the effectiveness of using diffusion models to train feature extractors. Zhang et al. [31] used the semantic correlation diffusion model, which can utilize unlabeled samples during the denoising process to consider the semantic correlation of spectral–spatial features, and retrieve the features of the original image semantic correlation to mitigate pseudo change between multitemporal images arising from inconsistent imaging conditions. Bandara et al. [30] pretrained DDPMs using unlabeled remote sensing data and used them as encoders to extract key features for single-modal CD.

Inspired by the success of diffusion models in generative artificial intelligence [59], Jia et al. [32] innovatively used conditional generative diffusion models to generate change maps by gradually denoising the pure noise, with differential features of bitemporal image as conditions. Wen et al. [33] designed a more complex generative CD model that generates change maps iteratively from pure noise by using pixel-level information and abstract features of the original images as conditions.

Overall, a lot of work has already demonstrated the effectiveness of diffusion models in the field of CD. However, the above-mentioned methods are all designed for single-modal CD, and their performance in MCD tasks remains to be verified. Therefore, it is of great significance and feasible to apply diffusion models to design a method for MCD.

III. Method

In an effort to tackle the formidable challenges presented by data heterogeneity in the realm of MCD, we put forward the D3PM model, which employs a dual-stream encoder structure with cross-modal feature alignment capabilities to process bitemporal images, and integrates multilevel features in a single decoder to predict change maps that achieved high-precision CD. Its overall architecture is illustrated in

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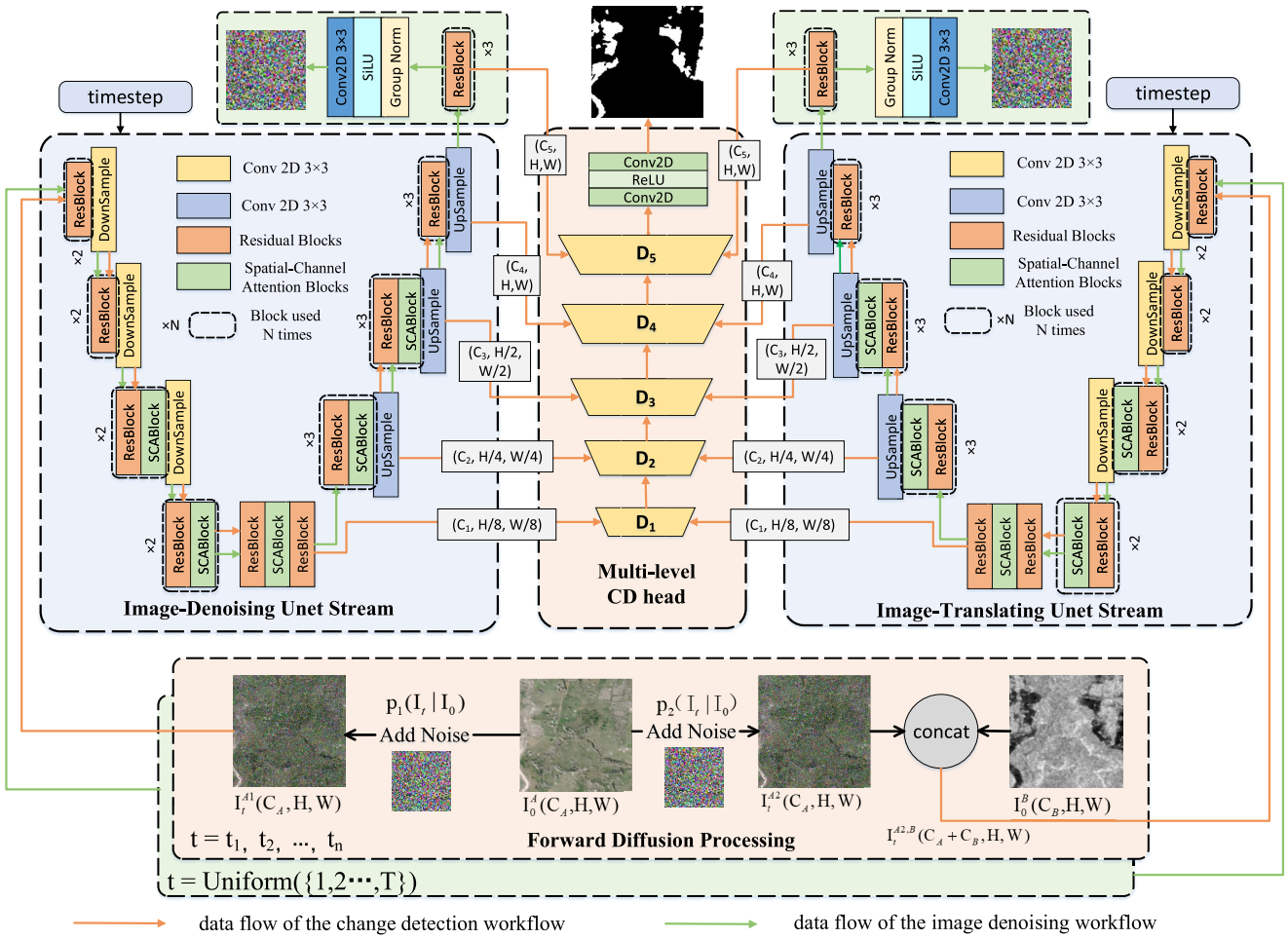


Fig. 1. Architecture of D3PM network mainly consists of three parts: (I) The IDUS receives the noisy image I_t^{A1} and timestep embedding corresponding to the forward process p_1 as input, and it is designed to extract features from image I_0^A when denoising I_t^{A1}. (II) According to the practice of Bai et al. [60], we incorporate SAR images as conditional constraints in the forward denoising process of optical images during the training process in order to achieve the translation from SAR-to-optical modality. The image translating U-Net stream receives the concatenated result of the noisy image I_t^{A2} and image I_0^B, along with the timestep embedding corresponding to p_2, and its function is to extract features from I_0^B during the translating process. (III) The CD head integrates the multilevel and multitimestep features extracted by the dual-stream U-Net. Due to the design of the ITUS, these features can be considered to have the same distributional properties, allowing for direct comparison and yielding change maps.

Fig. 1. In this section, we will first introduce the DDPM. Subsequently, we will provide a detailed description of the architecture of the D3PM model, which encompasses three main components: forward diffusion preprocessing, the dual-stream U-Net, and the multilevel feature fusion CD head. Finally, we will present the multitask collaborative training strategy.

A. Denoising Diffusion Probabilistic Model

As a generative model, DDPM was originally proposed to deal with the problem of image generation. Its sampling process is a Markov chain that reconstructs data samples through a step-by-step denoising process, beginning with randomly sampled Gaussian noise. The content below describes a brief overview of DDPM.

Typically, DDPM uses two Markov chains to learn the distribution $p(x_0)$ of training data: in the forward process, Gaussian noise is gradually added to the clean data sample $x_0 \sim p(x_0)$ until the data are entirely destroyed, resulting in an isotropic gaussian distribution $\mathcal{N}(0, I)$. The noising operation

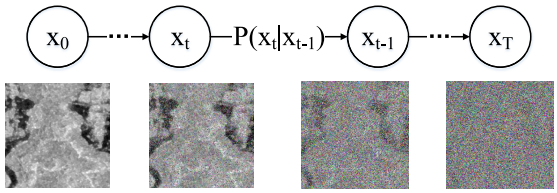


Fig. 2. Visual forward process for SAR image.

at each timestep $t \in \{0, 1, \ldots, T\}$ of the forward chain can be defined as follows:

$$p(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right) \tag{1}$$

where T is a predefined parameter, which is the maximum value of t, $(x_0, x_1, \ldots, x_t, \ldots, x_T)$ represents a T-step Markov chain, $(\beta_0, \beta_1, \ldots, \beta_t, \ldots, \beta_T) \in (0, 1)$ represent the noise schedule and β_t increases as the t increases. For image data, its visual forward process is shown in Fig. 2.

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After parameter renormalization, given a clean data sample x_0, a noisy data x_t can be directly obtained without needing iterative steps by sampling the Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ and applying the following transformation to x_0:

$$p(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, \left(1-\bar{\alpha}_t\right)I\right) \tag{2}$$

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon_t, \epsilon_t \sim \mathcal{N}(0, I) \tag{3}$$

where $\bar{\alpha}_t = \prod_{i=0}^{t} \alpha_i = \prod_{i=0}^{t} (1-\beta_i)$. Using (3), we can obtain x_t by adding noise to x_0 for each $t \in \{0, 1, \ldots, T\}$.

A U-Net [61] like structure network ϵ_θ is trained to predict ϵ that added to x_0 by minimizing the following loss function:

$$\begin{aligned}\mathcal{L} &= \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \\ &= \left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2.\end{aligned} \tag{4}$$

During the reverse process, given a noisy data x_t and its corresponding t, the trained network ϵ_θ can obtain x_{t-1} though predicting the noise that added to x_{t-1}. This stage is defined as follows:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z \tag{5}$$

$$\sigma_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t \tag{6}$$

where $z \sim \mathcal{N}(0, \mathrm{I})$. In the case where x_t represents pure noise sampled randomly, upon completion of the full reverse sampling process, i.e., $x_t \to x_{t-1} \to \cdots \to x_0$, the randomly generated data x_0 is derived.

As the original DDPM method is unconditional, the notation $\epsilon_\theta(x_t, t)$ in (4) and (5) indicates that no supplementary conditional information are fed into the network when it predicts the noise throughout both the training and sampling procedures. In the case of conditional diffusion models, an information y is fed into the network to guide the denoising process, this means that $\epsilon_\theta(x_t, t)$ is replaced by $\epsilon_\theta(x_t, y, t)$.

B. Forward Diffusion Preprocessing

The forward diffusion process can help eliminate the domain-specific features of input images and align the domains of diffused images, thus facilitating the extraction of semantics within the common domain [62]. The idea is that multimodal data belong to different domains, but their semantic contents should share the same domain. This inspires us to use the forward processing proposed in DDPM to weaken the differences in detail such as edges and textures caused by different modalities under the same semantic content of the input images.

Considering the multitask end-to-end training of D3PM, a special design for the timestep is required. We first follow the method of DDPM for training the network to perform denoising, that is the timestep t is randomly obtained in $\{0, 1, \ldots, T\}$. Meanwhile, compared with random steps, combining the feature representations of noisy images from multiple certain diffusion steps can significantly improve the accuracy of dense prediction tasks [27]. Therefore, we also design multiple noisy versions of prechange and postchange images and obtain corresponding multitimesteps features. Here, we use (3) to corrupt the input remote sensing image I_0^A at timestep $t \in \{\text{Uniform}(1, 2, \ldots, T), t_1, t_2, \ldots, t_n\}$ to obtain I_t^{A1} and I_t^{A2}

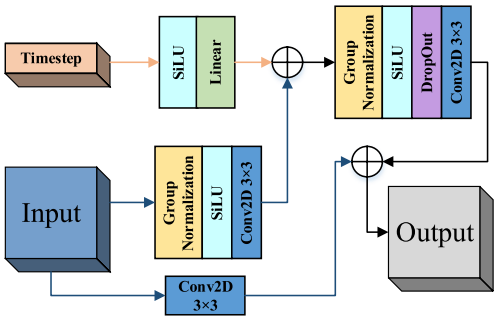


Fig. 3. Residual blocks mainly consist of linear layers, SiLU activation function, group normalization, 2-D convolutional layers, and dropout activation function.

$$I_t^{A1} = I_0^A\sqrt{\bar{\alpha}_t} + \epsilon_t^1\sqrt{1-\bar{\alpha}_t}, \quad \epsilon_t^1 \sim \mathcal{N}(0, I) \tag{7}$$

$$I_t^{A2} = I_0^A\sqrt{\bar{\alpha}_t} + \epsilon_t^2\sqrt{1-\bar{\alpha}_t}, \quad \epsilon_t^2 \sim \mathcal{N}(0, I). \tag{8}$$

Notably, under the same timestep t, we employ two forward diffusion processes to obtain two different noisy images I_t^{A1} and I_t^{A2} by utilizing the randomness of Gaussian noise ϵ. This is because we found in practice that adding the same noise to two images would make the input and output of the two-stream network highly similar, which may lead to convergence problems and instability during the training process, thereby reducing the accuracy of CD. After completing the forward diffusion process, I_t^{A1} and t are directly input into the IDUS and extract multiscale and multitimestep features of I_0^A, which are used to predict the noise ϵ_t^1 (where t is uniformly sampled from $\{1, 2, \ldots, T\}$) and the CD results (for specific $t \in \{t_1, t_2, \ldots, t_n\}$). I_0^B and I_t^{A2} are concatenated along the channel dimension to obtain $I_t^{A2,B}$, and then together with t, are input into the ITUS, which is designed for modality translation and predicts ϵ_t^2 and extract features of I_0^B. As mentioned above, we are less concerned with the intuitive results of image translation and instead view the translating operation as a feature extraction process for assisting MCD, during which the features of I_0^B and I_0^A learned are meant to reduce modality heterogeneity, thereby facilitating the comparison to obtain the change map.

C. Image Denoising and ITUS

The architecture of the network used to predict noise is based on U-Net, its detailed structure is shown in Fig. 1, which is primarily composed of three parts as follows.

1) Residual Block: As Fig. 3 shows, it takes feature maps from the previous layer of the network and timestep embedding as input and then outputs the result of shortcut connections or skip connections. Due to the large scale of our dual-stream network, residual blocks are essential for addressing the vanishing and exploding gradient problems,

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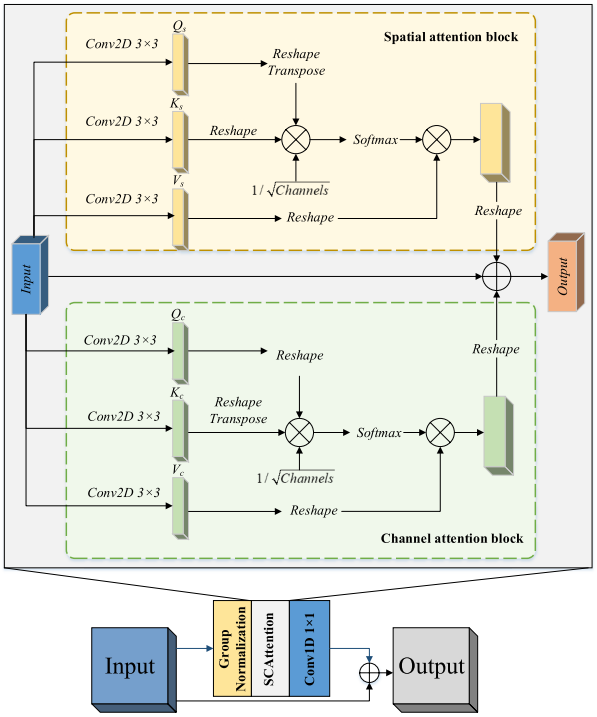


Fig. 4. Simplified structure of SCAblock. It contains group normalization, 2-D convolutional layers, and spatial–channel attention workflow denoted by SCAttention.

which help stabilize the deep neural network during training and accelerate the convergence of the model.

2) Spatial–Channel Attention Block: Considering the specificity of remote sensing images, we specifically designed two types of attention mechanisms to fully leverage the dependencies between pixels from both spatial and channel perspectives and combined the results through weighted summation. The simplified structure of SCAblock and the workflow of spatial–channel attention mechanism (SCAttention) are shown in Fig. 4. SCAblock takes feature maps from residual blocks as input and then outputs results with an enhanced focus on spatial resolution and channel context information.

For spatial–channel attention, specifically, given an input feature $X \in \mathbb{R}^{C\times H\times W}$ (where C is the number of channels, and H and W are the height and width of the feature maps), spatial attention calculates query (Q_s), key (K_s), and value (V_s) using three 2-D convolutional layers with 3×3 kernels. Q_s, K_s, and V_s matrices have the same dimensions as X. Next, the Q_s, K_s, and V_s matrices are reshaped to $C \times L$ (where $L = H \times W$), and then, we can obtain the spatial attention result through the following formula:

$$\text{Spatial Attention} = \text{Reshape}\left(V_s\left[\text{SoftMax}\left(\frac{Q_s^T K_s}{\sqrt{C}}\right)\right]\right) \quad (9)$$

where spatial attention $\in \mathbb{R}^{C\times H\times W}$.

The workflow of channel attention is similar to that of spatial attention, but the main difference here is reshaping and multiplying Q_c and K_c to obtain a channel attention matrix with dimensions $C \times C$. Then, through the same operations, we can get a channel attention result $\in \mathbb{R}^{C\times H\times W}$

$$\text{Channel Attention} = \text{Reshape}\left(\left[\text{SoftMax}\left(\frac{Q_c K_c^T}{\sqrt{C}}\right)\right]V_c\right). \quad (10)$$

Finally, adding the original feature map X and the two attention results yields the output of spatial–channel attention, as shown in the following equation:

$$\text{Output} = \text{Channel Attention} + \text{Spatial Attention} + X. \quad (11)$$

3) Downsample and Upsample: In order to reduce the computational complexity and integrate multiscale features, we conduct downsampling and upsampling operations on the feature maps. These operations are accomplished by 2-D convolutional layers. For example, given an original input with a resolution of $H \times W$, after passing through a downsampling or upsampling module, the feature map resolution becomes $H/2 \times W/2$ or $H*2 \times W*2$, respectively. The resolution of feature maps significantly impacts model performance in neural networks, so the size of the smallest feature map obtained after several downsampling operations should not be too large, as this can increase computational costs and lead to overfitting; conversely, a size that is too small can cause the feature map to lose valuable reference information, resulting in a loss of semantic details in the raw image. We use three downsampling operations, and with an input image size of 256×256, the smallest feature map resolution is 32×32, which is appropriate for the model. To balance computational cost and model performance, we select feature maps with the following dimensions as multiscale features: $192 \times 32 \times 32$, $192 \times 64 \times 64$, $144 \times 128 \times 128$, $96 \times 256 \times 256$, and $48 \times 256 \times 256$, which are then fed into the CD head for predicting change maps, as indicated in Fig. 1.

D. Noise Predictor

According to (3), by setting $t = \text{Uni}(1, 2, \ldots, T)$, we can obtain the data I_t^{A1} and $I_t^{A2,B}$ augmented with noise ϵ_t^1 and ϵ_t^2 at stochastic timesteps. By inputting I_t^{A1} and $I_t^{A2,B}$, the model is enabled to learn the intrinsic data features in the denoising task, and the features output by the U-Net network can be represented as F_1^d and F_2^d

$$F_d^1 = F_{t=\text{Uni}(1,2,\ldots,T)}, \quad F_d^2 = F_{t=\text{Uni}(1,2,\ldots,T)}. \quad (12)$$

where Uni denotes the random selection of a number from 1 to T. F_d^1 and F_d^2 are first normalized by group normalization and then activated by the SiLU function. They are subsequently fed into a 2-D convolutional layer, respectively, to obtain final prediction results that closely approximate ϵ_t^1 and ϵ_t^2.

E. CD Head

Unlike random timesteps, we input data augmented with noise at fixed time intervals to train the model on the CD task. In addition, feature maps of multiple scales are extracted to enable the model to simultaneously focus on both low-level features such as edges, textures, and colors, and high-level features like objects, scenes, and contextual information.

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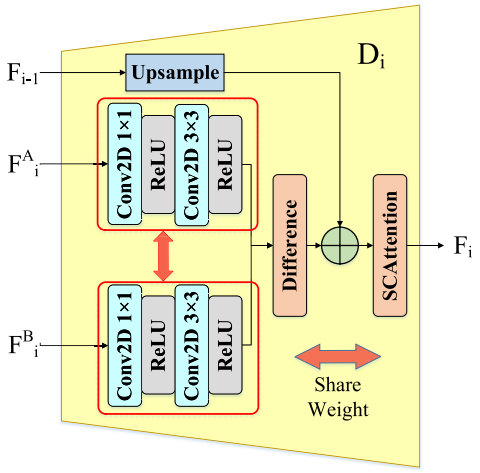


Fig. 5. CD head employs a multilevel decoding layers design, and the image above shows one of the layers. It should be emphasized that D_1 and D_5 do not have upsample modules.

As shown in Fig. 5, we designed a hierarchical CD head consisting of five decoder blocks, denoted by D_i, where $i \in \{1, 2, \ldots, 5\}$. At ith spatial scale, the fused multitimestep features of I^A and I^B is marked as F_i^A and F_i^B, respectively,

$$F_i^A = \text{Cat}\left(F_{i,t=t_1}^A, F_{i,t=t_2}^A, \ldots, F_{i,t=t_n}^A\right) \tag{13}$$

$$F_i^B = \text{Cat}\left(F_{i,t=t_1}^B, F_{i,t=t_2}^B, \ldots, F_{i,t=t_n}^B\right) \tag{14}$$

where Cat represents the operation of channel dimension concatenation.

D_i is responsible for processing and fusing F_i^A and F_i^B at ith spatial scale and F_{i-1} at $(i-1)$th spatial scale, respectively. It applied convolutional layers to align the number of channels for features at different spatial scales and introduced nonlinear relationships to improve the learning ability and training effect of the model. Specifically, both F_i^A and F_i^B undergo two 2-D convolutional layers with ReLU activation in between and at the end, and then the absolute difference between the two results is calculated to obtain a semantic representation of change regions

$$F_i^{\text{diff}} = \left|\text{net}\left(F_i^A\right) - \text{net}\left(F_i^B\right)\right| \tag{15}$$

where F_i^{diff} is the difference result and net represents the convolution and activation operations.

F_{i-1} is upsampled using bilinear interpolation to match the resolution of F_i^A and F_i^B. The two matrices are then added elementwise and enhanced with SCAttention to improve the focus of the network on channel and spatial information, as depicted in the following expression:

$$F_i = \text{SCAttention}\left(\text{Upsample}(F_{i-1}) + F_i^{\text{diff}}\right). \tag{16}$$

Note that D_5 does not require upsampling features from D_4 because the sizes of F_4 and F_5 are the same. Finally, the feature map output by D_5, which has the same sizes as the original image pairs (i.e., $H \times W$), is fed into two convolutional layers with activation function in between to obtain the final binary classification prediction map, which has a dimension of $2 \times H \times W$.

F. Training

In the context of developing a multitask learning framework, we formulated a collaborative learning optimization strategy with asynchronous time steps for D3PM, which means that the input consists of a combination of data with noise added at multiple distinct timesteps, and the network is optimized simultaneously for both denoising and CD tasks. D3PM undertakes supervised training by leveraging labeled bitemporal multimodal data, and the overall loss function is configured as a weighted sum of the denoising loss and the CDloss, which is formalized as follows:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^{3} \lambda_i \cdot f_i(p_i, l_i) \tag{17}$$

$$\sum_{i=1}^{3} \lambda_i = 1 \tag{18}$$

here $i \in \{1, 2, 3\}$, λ_i represents the weight of the loss function f_i, p_i represents predictions of noise and CD results output by the network, and l_i signifies the corresponding ground truth.

Diffusion models can learn robust image features during the denoising process [34]. To fully leverage this, we designed the denoising loss for a dual-stream network using mean absolute error (MAE) loss

$$f_{1,2} = \text{MAE}(\epsilon_k, \hat{\epsilon}_k) = \frac{1}{n * N} \sum_{k=1}^{n*N} |\epsilon_k - \hat{\epsilon}_k| \tag{19}$$

where $\hat{\epsilon}_k$ is the prediction of the network, ϵ_k is the true noise added to the image, n refers to the number of noisy image versions obtained by performing multiple forward diffusions on an original image, and N represents the batch size of training samples.

The CD task involves pixelwise prediction of changes at each location; therefore, we use the cross-entropy loss (CE loss) function to define the CD loss

$$f_3 = \text{CE}(y_j, \hat{y}_j) = -\frac{1}{N * L} \sum_{j=1}^{N*L} \left(y_j \log(\hat{y}_j) + (1 - y_j) \log(1 - \hat{y}_j)\right) \tag{20}$$

where $y_j \in \{0, 1\}$ represents the ground truth, 0 represents no change, 1 represents a change, and $\hat{y}_j \in (0, 1)$ is the probability value that the pixels predicted for the network belong to the two categories. L is the number of pixels in change maps (i.e., $H \times W$); thus, $N \times L$ represents the total number of samples in a batch that need to be classified.

By simultaneously optimizing the aforementioned MAE loss and CE loss, the unconditional DDPM branch is enabled to extract noise-robust features across multiple timesteps for the denoising task. Meanwhile, the conditional DDPM branch can reduce the heterogeneity of features and enhance the comparability of multimodal data features through modality transformation, thereby facilitating CD.

IV. Experiments

In this section, we conduct experiments and analysis on the proposed D3PM. First, we introduce the datasets used in

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the experiment. Next, we list the evaluation metrics used for the results, as well as our experimental settings of hardware and software. Finally, we present and analyze the quantitative and visual comparison results of this proposed method with others, as well as the ablation study results involving modules and hyperparameters.

A. Datasets

In this experiment, we selected two representative MCD datasets to demonstrate the advantages of our proposed D3PM, which is detailed below.

1) CAU-Flood: The CAU-Flood dataset [50] collected pre-disaster Sentinel-2 and post-disaster Sentinel-1 data from 18 flood-affected areas in China, Bangladesh, Australia, the United States, Canada, and Germany, its total coverage area is 95 142 km^2. It aims to detect areas inundated by floods (i.e., change areas before and after disasters) when post-disaster Sentinel-2 data is unavailable. Compared to single-modal CD datasets, this dataset uses both SAR and optical images, making the task more challenging and requiring more robust feature information to obtain a change map. It is worth noting that the SAR images only have data in the VV-polarization mode, while the optical images include red, green, blue, and near-infrared bands, with both having a resolution of 256 × 256. The dataset is divided into trainset and testing sets at a 5:1 ratio, with 15 231 image pairs contained in the training set and 3071 pairs in the testing set.

2) MT-Wuhan: The MT-Wuhan dataset is collected over the urban area of Wuhan, China, and includes: a Sentinel-2 optical image acquired in March 2017 with a resolution of 10 m, and a SAR amplitude product derived from an HH-polarized image acquired by the COSMO-SkyMed constellation in StripMap HIMAGE mode in March 2020 with a ground resolution of 3 m. The change label is obtained by manual region of interest (ROI) labeling using ENVI software. The size of the original image is 11 216 × 13 693, and preprocessing of this dataset includes radiometric and geometric correction, clipping, and log transformation, to allow the statistical distribution characteristics of the SAR image to be similar to an RGB image [63]. Bilinear interpolation upsampling is conducted on the low-resolution optical image to make its spatial resolution the same as the SAR image. The ratio of changed and unchanged pixels is 1:5.

B. Experimental Setup

Below, we describe the criteria used to evaluate the performance of different methods and the implementation details of D3PM for experiments.

1) Evaluation Criteria: In order to quantitatively evaluate the performance of the methods in the experiments, several credible evaluation criteria best suited for each dataset are employed.

Cause the CD results are binary, we first calculate the confusion matrix of the results. As shown in Table I, it involves four indicators: true positive (TP), true negative (TN), false positive (FP), and false negative (FN), which separately define the number of pixels that are correctly classified as a change, correctly classified as no change, misclassified as a change, and misclassified as no change. Based on this confusion matrix, the following indices can be calculated as follows.

TABLE I
CONFUSION MATRIX

Actural\Rredicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

1) *Mean Intersection Over Union (mIoU):*

$$\text{mIoU} = \frac{1}{2} \cdot (\text{CIoU} + \text{UCIoU}) \tag{21}$$

where

$$\text{CIoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{22}$$

and

$$\text{UCIoU} = \frac{\text{TN}}{\text{TN} + \text{FP} + \text{FN}}. \tag{23}$$

CIoU and UCIoU represent the IoU of a change and no change, respectively.

2) *F1 Score, Precision, and Recall:*

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{24}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{25}$$

$F1$ can achieve a balance between precision and recall in the field of classification tasks

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{26}$$

2) Implementation Details: D3PM was implemented based on the PyTorch framework [64] and trained on one Nvidia RTX 3090 GPU. During the training process, we use the Adam optimizer with an initial learning rate set to $1e^{-4}$, and the exponential decay rates of the first and second moments of the gradient are set to 0.5 and 0.999, respectively. We employed mini-batch training with a batch size of 2, primarily due to the large data volume and limited memory. To enhance training quality and model generalization, we implemented two data augmentation strategies: random rotation and horizontal flipping. Each training sample was rotated 90° and flipped horizontally with an independent probability of 0.5. The model converges at different rates on the CAU-Flood and MT-Wuhan datasets, so 50 and 200 epochs are set for them, respectively. Finally, we tested the model after each training epoch to monitor the optimal performance that the model can achieve.

C. Comparative Experiments

1) Comparison Methods: In this experiment, we selected several state-of-the-art models for comparison to prove the superiority of our proposed D3PM. Fully convolutional early-fusion (FC-EF) [38], fully convolutional Siamese difference (FC-Siam-diff) [38], fully convolutional Siamese concatenation (FC-Siam-conc) [38], and Satellite-Uav Network (SUNet)

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TABLE II
QUANTITATIVE TESTING RESULTS OF VARIOUS METHODS ON CAU-FLOOD AND MT-WUHAN DATASETS

Backbone	Method	CAU-Flood				MT-Wuhan			
		F1	mIoU	Precision	Recall	F1	mIoU	Precision	Recall
CNN	(a) FC-EF (2018)	0.8863	0.8794	0.8679	0.9056	0.5539	0.6255	0.5663	0.5421
	(b) FC-Siam-Diff (2018)	0.5514	0.6411	0.8485	0.4083	0.5200	0.6084	0.5593	0.4859
	(c) FC-Siam-Conc (2018)	0.8293	0.8312	**0.9806**	0.7184	0.5439	0.6217	0.5761	0.5152
	(d) SUNet (2021)	0.7486	0.7456	0.6041	**0.9840**	0.2869	0.2786	0.1757	**0.7819**
	(e) SNUNet (2021)	0.8883	0.8829	0.9441	0.8388	0.4439	0.6221	0.4274	0.4618
	(f) DA-MSCDNet(2022)	0.9081	0.9009	0.8954	0.9212	0.5443	0.6246	0.6040	0.4953
	(g) ELGCNet(2024)	0.9000	0.8929	0.8852	0.9154	0.4577	0.5533	0.4021	0.5311
Transformer	(h) ChangeFormer(2022)	0.9011	0.8941	0.8958	0.9066	0.5478	0.6172	0.5281	0.5690
	(i) SwinSUnet (2022)	0.9087	0.9016	0.8995	0.9180	0.5136	0.5973	0.5001	0.5279
	(j) ScratchFormer (2024)	0.8862	0.8795	0.8775	0.8950	0.4647	0.5629	0.4268	0.5099
	(k) CMMAN (2024)	0.9087	0.9017	0.9052	0.9123	0.5436	0.6238	0.5988	0.4976
	(l) MMCD (2024)	0.8919	0.8847	0.8707	0.9141	0.4537	0.5707	0.4870	0.4246
Diffusion model	(m) DDPM-CD (2024)	0.8970	0.8893	0.8598	0.9375	0.5383	0.6234	0.5740	0.5068
	(ours)D3PM	**0.9143**	**0.9070**	0.9090	0.9197	**0.5675**	**0.6387**	**0.6418**	0.5086

[18] adopt a Siamese network design based on CNNs; Siamese network and Nested U-Net (SNUNet) [40]; and efficient global and local context aggregation network (ELGCNet) [39] introduce attention mechanisms into the Siamese CNN; ChangeFormer [43], ScratchFormer [44], and SwinSUnet [46] MMCD [65] methods design a Siamese network based on transformers. Comparing D3PM with methods that also adopt Siamese network design can prove the effectiveness of our use of two diffusion processes to learn image features, and the comparison results with DA-MSCDNet [17], SUNet and CMMAN [58] methods which are specifically designed for MCD can prove the effectiveness of our translation strategy. Finally, we also compared with the DDPM-CD [30] method, which also uses DDPM as the backbone.

To control variables and obtain fair results, the experimental settings of the comparison methods will be kept the same as those of D3PM. It should be noted that due to the randomness of the noise added in the forward process [as shown in (3)], there will be slight differences between the input and output of the network. Hence, we tested the D3PM method five times and took the average of the five results as the final testing outcome. As shown in Table II, the evaluation metrics of each algorithm on the two testing sets are presented, and the highest score is indicated in bold font and the second highest is underlined.

2) Results on the CAU-Flood Dataset: The CAU-Flood dataset covers a wide geographical area and is composed of optical and SAR image pairs for flood detection. Due to the differences in sensor imaging mechanisms, optical and SAR data have a high level of image heterogeneity. Processing these data is a significant challenge for multimodal CD algorithms. Despite these difficulties, D3PM has shown excellent performance on the CAU-Flood dataset. The testing results in Table II show that D3PM has achieved an impressive $F1$ score of 0.9143 and mIoU of 0.9070, surpassing other methods. Some methods achieve higher precision or recall values, such as FC-Siam-Conc (precision of 0.9806 and recall of 0.7184) and SUNet (precision of 0.6041 and recall of 0.9840). However, since they also have more FNs or FPs, they are unable to obtain high Precision and Recall simultaneously, which also leads to lower $F1$ scores and mIoU, so their overall performance is not the best. Although D3PM does not achieve the highest precision and recall, it does not suffer from an imbalance between these two metrics. D3PM can accurately predict TNs, and TPs and significantly reduce the number of FNs and FPs. This highlights that D3PM can effectively weaken the substantial differences caused by imaging modes and accurately detect changes in flood areas in different geographical environments.

The visual results in Fig. 6 can intuitively judge the superiority of D3PM. The detection map it generates indicates its excellent ability to distinguish flood-affected areas and unaffected areas. This ability has crucial guiding significance for planning rescue and post-disaster assessment. However, the visualization results of other methods show that most of them cannot predict detailed segmentation boundaries and continuous change areas.

3) Results on the MT-Wuhan Dataset: The MT-Wuhan dataset is composed of image pairs from COSMO-SkyMed and Sentinel-2. The D3PM has the highest $F1$ (0.5675), indicating that it achieves the best balance between precision and recall. The mIoU (0.6387) suggesting that D3PM can align features well, thereby detecting change areas more accurately. Although the precision (0.6418) and recall (0.5086) suggest that D3PM predicts many FNs, this does not hinder its overall performance from being the best. These results prove the robustness of D3PM in handling the heterogeneity introduced by different imaging platforms and environmental conditions.

As can be seen from Fig. 6, D3PM effectively captures the complex spatial relationships in urban environments, and its detection results clearly and accurately depict the changed

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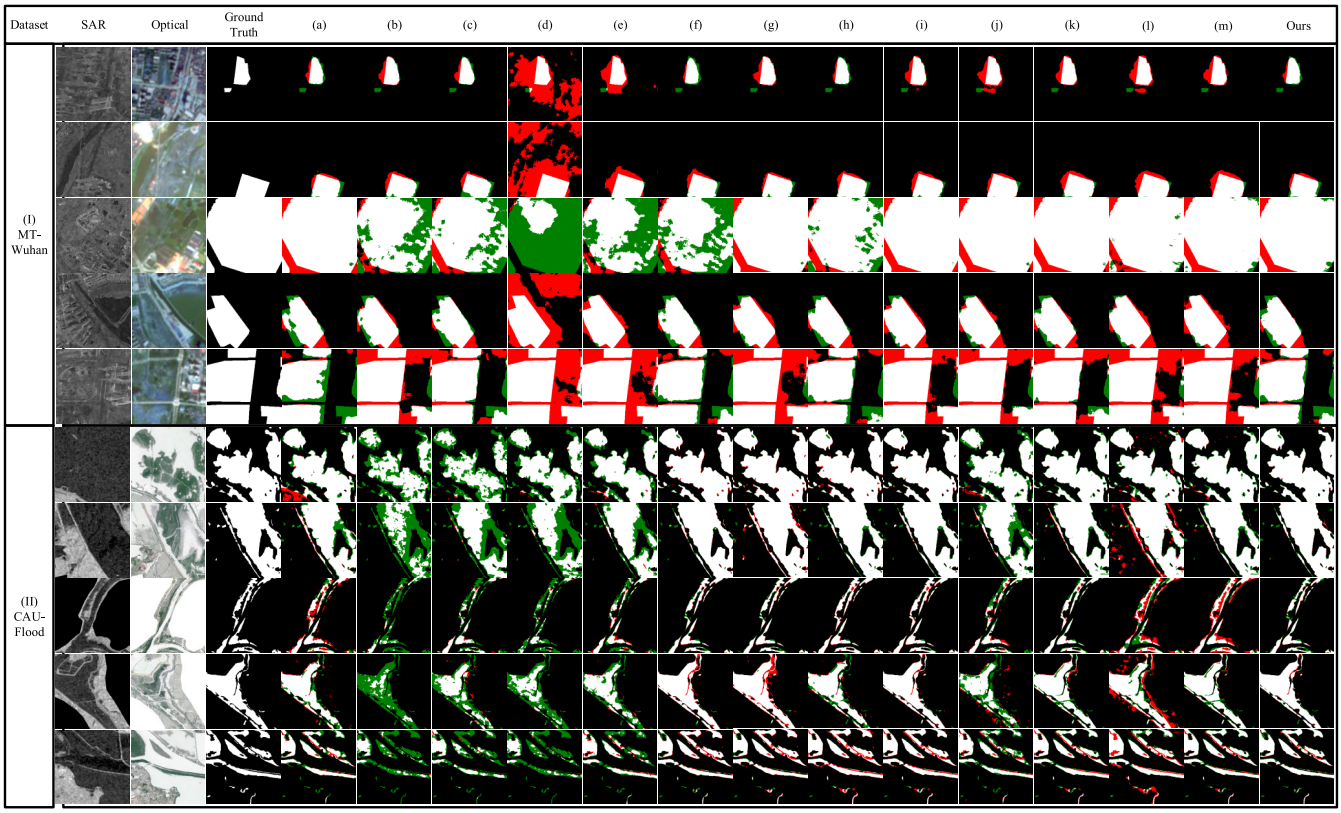


Fig. 6. Visualization results of various methods on CAU-Flood and MT-Wuhan datasets, and the white, black, green, and red pixels represent TPs, TNs, FNs, and FPs, respectively. (a) FC-EF, (b) FC-siam-diff, (c) FC-siam-conc, (d) SUNet, (e) SNUNet, (f) DA-MSCDNet, (g) ELGCNet, (h) ChangeFormer, (i) SwinSUnet, (j) ScratchFormer, (k) CMMAN, (l) MMCD, (m) DDPM-CD, and (ours) D3PM.

areas. Compared with other methods, its FPs and FNs are significantly reduced.

D. Parameter Analysis and Ablation Study

The purpose of this parameter ablation experiment is to investigate the effectiveness of the D3PM architecture design and to explore the impact of some important hyperparameters on the testing results. In Tables III–V, we marked the highest score in bold and the second-highest score with underline. It is worth stating that the experimental group with bold-faced font corresponds to our final settings. Based on these final settings, the following experiments are conducted in accordance with the principle of a single variable.

1) Analysis of Timestep: To determine the most suitable timestep t, we studied the impact of different values of t on CD tasks during the forward diffusion process. According to (2) and (3), a larger value of t will lead to a higher proportion of noise added to the image, causing more serious interference to the raw information; on the contrary, a smaller value of t can enable the image to retain more of its original information. Besides, fusing the features of multiple noisy versions of the original image will help obtain a more accurate prediction map. At the same time, in order to illustrate that the model can learn semantic features at specific diffusion steps $t_1, t_2, \ldots, t_n$, we set up an experimental group that uses three random steps. Table III shows the testing results of each group. Fig. 7 and Table III show the testing scores and visualization results for each group, respectively.

TABLE III

EXPERIMENTAL RESULTS WHEN t TAKES DIFFERENT VALUES. IN THIS TABLE, $i = 1, 2$, AND 3

Groups	CAU-Flood				MT-Wuhan			
	F1	mIoU	Precision	Recall	F1	mIoU	Precision	Recall
Random t	0.8979	0.8906	0.8777	0.9190	0.5202	0.6155	0.6413	0.4376
t=50	0.9043	0.8972	0.8939	0.9150	0.5301	0.6182	0.6109	0.4682
t=150	0.9027	0.8953	0.8796	**0.9270**	0.5281	0.6188	0.6310	0.4542
t=250	0.9034	0.8960	0.8941	0.9129	0.5173	0.6176	**0.6991**	0.4105
Random t_i	0.9083	0.9013	0.9049	0.9118	0.5327	0.6127	0.6783	0.4385
t_i=25,50,75	0.9119	0.9050	**0.9167**	0.9072	**0.5694**	**0.6416**	0.6535	0.5044
t_i=50,150,250	**0.9143**	**0.9070**	0.9090	0.9197	0.5675	0.6387	0.6418	**0.5086**
t_i=50,100,400	0.9058	0.8987	0.8976	0.9141	0.5545	0.6310	0.6172	0.5034

The results show that multiple determined diffusion steps make the model perform better than random steps. Although adding noise will inevitably interfere with the original data, we believe that when t is set to appropriate values, the corresponding noisy versions of the original image can weaken the limitations imposed by the differences between modalities on CD. At the same time, it will not excessively damage the original data and thus retain the important semantic features required by the network.

2) Analysis of Weighted Sum Loss Function: D3PM represents a multitask collaborative optimization framework that integrates the denoising task and the CD task; therefore, the setting of the loss function will exert an impact on the optimization of the model. The overall loss function, as (17)

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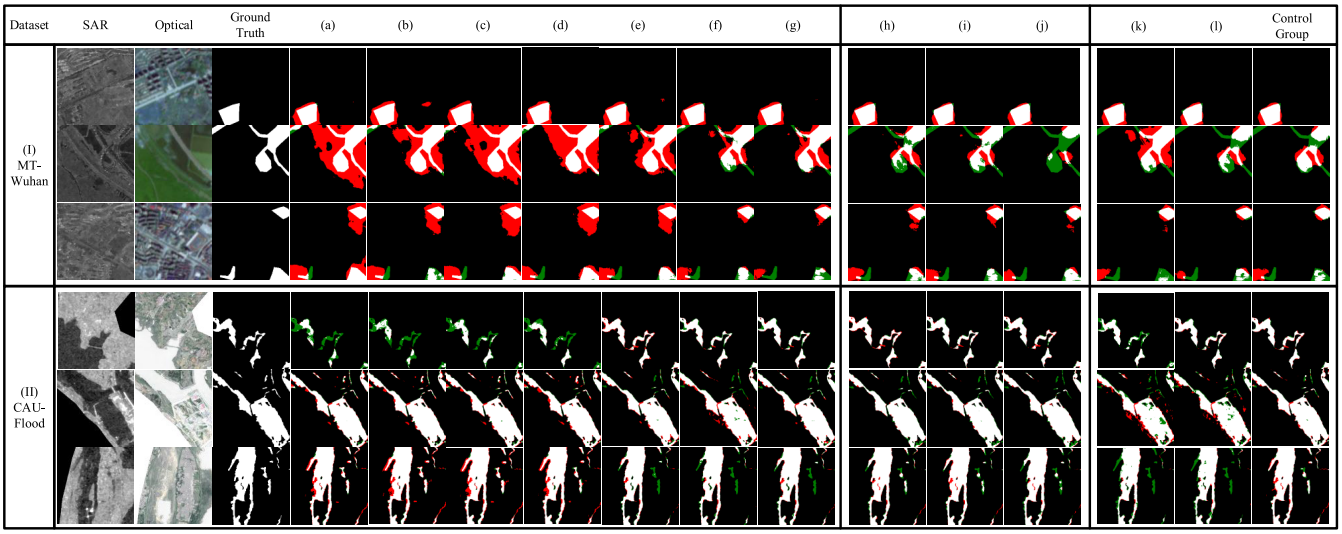


Fig. 7. Visualization results of various experimental groups on CAU-Flood and MT-Wuhan datasets. (a) Random t, (b) $t = 50$, (c) $t = 150$, (d) $t = 250$, (e) random $t_{1,2,3}$, (f) $t_{1,2,3} = 25$, 50, and 75, (g) $t_{1,2,3} = 50$, 100, and 400, (h) $\lambda_{1,2,3} = 0.25$, 0.25, and 0.50, (i) $\lambda_{1,2,3} = 0.20$, 0.20, and 0.60, (j) $\lambda_{1,2,3} = 0.10$, 0.10, and 0.80, (k) notranslation, (l) OPT2SAR. The control group represents the final experimental setting, i.e., $t_{1,2,3} = 50$, 150, and 250, $\lambda_{1,2,3} = 0.15$, 0.15, and 0.70 as well as SAR images inputting ITUS and optical images inputting IDUS.

TABLE IV

EXPERIMENTAL RESULTS WHEN $\lambda_{1,2,3}$ TAKES DIFFERENT VALUES

Groups	CAU-Flood				MT-Wuhan			
	F1	mIoU	Precision	Recall	F1	mIoU	Precision	Recall
0.25,0.25,0.50	0.9105	0.9036	0.9138	0.9073	0.5593	0.6347	0.6323	0.5015
0.20,0.20,0.60	0.9134	0.9065	**0.9151**	0.9118	0.5599	0.6366	**0.6538**	0.4895
0.15,0.15,0.70	**0.9143**	**0.9070**	0.9090	0.9197	**0.5675**	**0.6387**	0.6418	**0.5086**
0.10,0.10,0.80	0.9037	0.8964	0.8820	**0.9266**	0.5455	0.6176	0.5892	0.5078

TABLE V

EXPERIMENTAL RESULTS UNDER DIFFERENT SETTINGS OF THE DENOISING AND TRANSLATING STREAM

Groups	CAU-Flood				MT-Wuhan			
	F1	mIoU	Precision	Recall	F1	mIoU	Precision	Recall
NoTranslation	0.8910	0.8848	**0.9137**	0.8694	0.5211	0.6146	0.6200	0.4495
OPT2SAR	0.9078	0.9006	0.8928	**0.9234**	0.5430	0.6246	0.6136	0.4870
SAR2OPT	**0.9143**	**0.9070**	0.9090	0.9197	**0.5675**	**0.6387**	**0.6418**	**0.5086**

shows, is the weighted sum of denoising loss and CD loss, and λ_i is the weight value for each function f_i. The optimization direction of the model during training changes with different weight values. Under the constraint that $\sum_{i=1}^{3} \lambda_i = 1$, if λ_1 and λ_2 are set to higher values, the model will be optimized to better predict noise added to noisy images; conversely, if λ_3 is set to a higher value, the model will optimize toward predicting more accurate change maps. To explore the most suitable weight values, we set up experimental groups with different sets of λ_i values. The results are as shown in Fig. 7 and Table IV.

Considering the two datasets comprehensively, when λ_i is set to $\lambda_{1,2,3} = 0.15$, 0.15, and 0.70, the quantified results are the best. Although intuitively a larger λ_3 should be beneficial for CD tasks, in fact, the results suggest that putting too much emphasis on optimizing the model for CD tasks did not always yield the best testing results, on the contrary, optimizing the model to predict accurate noise can also be beneficial for obtaining more accurate change maps. This indicates that features extracted for denoising tasks do not necessarily hinder CD tasks and may even improve the model's performance to some extent.

3) Analysis of Denoising and Translating Stream: Since the design of a dual-stream architecture with a denoising stream and a translating stream for multimodal processing of D3PM, it is necessary to determine which image in an input image pair is more suitable for the denoising stream while the other for the translating stream. Taking the CAU-Flood dataset as an example, a training sample consists of two images: a SAR image and an optical image. Optical images contain color and spectral information of the ground, which are beneficial for land cover segmentation and CD but it is susceptible to interference from weather and lighting conditions. SAR images, benefiting from their imaging mechanism, are not affected by weather and have high resolution, while containing rich land cover information. However, SAR images often have much noise that is difficult to handle. Therefore, in this ablation study, we attempted to convert SAR images to optical images (SAR2OPT) or optical images to SAR images (OPT2SAR) to obtain test metrics. In addition, to demonstrate the effectiveness of translating streams, we let both streams as denoising streams, meaning that noisy SAR and optical images were directly input into the network (noted as NoTranslation). The results are shown in Fig. 7 and Table V.

The results validate the effectiveness of our designed translation stream, as NoTranslation performs worse than OPT2SAR and SAR2OPT. In addition, the metrics and visualization results indicate that translating SAR images to optical images yields the best results. Our intuition is that the translation process may reduce noise in SAR images while enhancing features and information that are beneficial for detecting change regions, which allows the extracted SAR

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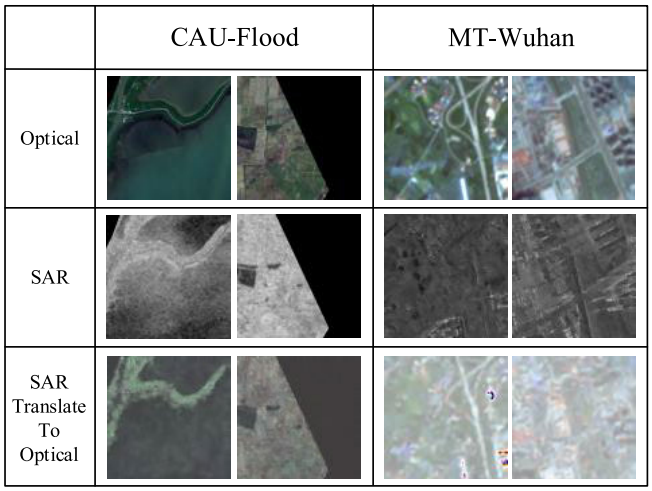


Fig. 8. Visual translation results of samples taken from CAU-Flood and MT-Wuhan datasets.

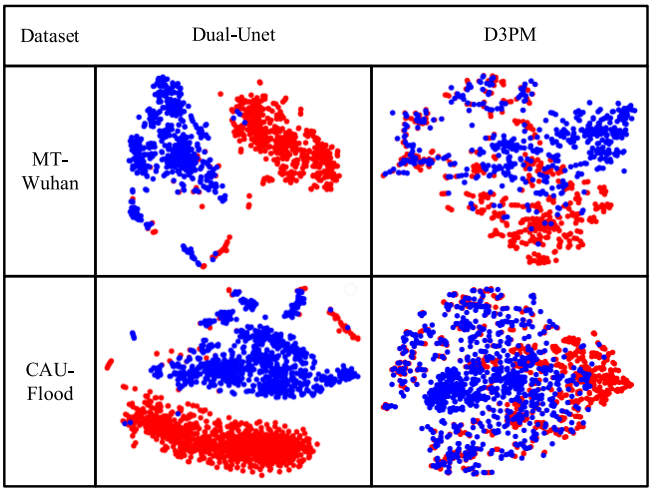


Fig. 9. Feature visualization analysis via t-SNE. The red and blue points represent features from two different modal images.

images semantic features to be directly compared with those of optical images, leading to accurate change prediction maps.

E. Interpretability Analysis

In order to verify that the proposed method can effectively reduce the heterogeneity among data and thus achieve the comparability of multimodal images, we have conducted interpretability analysis in this experiment from both the pixel level and the abstract feature level, respectively.

1) Analysis of the Image Translation Visual Results: The design intention of ITUS is to align multimodal data. During the inference process, instead of retraining the ITUS to complete image translation, we used the ITUS that had already been trained in the CD task to infer optical images. A SAR image was spliced with randomly sampled normal noise and then input into the network. According to (5), the optical image corresponding to the SAR image was gradually generated step by step from $T, T-1, \ldots,$ to 0, where $T = 500$. The results obtained from the inference are shown in Fig. 8. It can be seen that ITUS can capture distinct boundaries and contours, which are sufficient to distinguish ground objects. Since we did not specifically optimize the network to obtain better visual translation results, the obtained images have lost some details and there are color differences compared with the real optical images.

2) Analysis of Feature Alignment: We utilized t-distributed stochastic neighbor embedding (t-SNE) [66] to perform dimensionality reduction and visualization on the features extracted by the network from two images of different modalities, in order to intuitively verify the effectiveness of the proposed feature alignment method. The Dual-U-Net method adopts a network that is completely identical to that of D3PM. However, during the training process, the Dual-U-Net directly inputs two original images into two-stream networks without employing both conditional and unconditional diffusion processes to process the images. For D3PM, the multitimestep features output from the last upsample module of the U-Net encoder are extracted and concatenated along the channel dimension; for dual-U-Net, the features output from the same layer are extracted and replicated three times along the channel dimension. We cluster these features extracted from the images by the U-Net of the two methods, respectively, using the t-SNE algorithm. The results are shown in Fig. 9, where the red and purple dots represent the features of the two modal images, and the degree of aggregation or separation of the clusters of different colors indicates the similarity or difference in the feature distribution. Instead of using the ITUS and IDUS, the dual-U-Net model with the direct input of two original images yields results that show a clear separation, demonstrating a significant difference in the feature distribution; the results obtained from the D3PM model, on the other hand, significantly reduces the feature distribution differences caused by modal heterogeneity, thereby providing strong visual evidence of the method's effectiveness.

F. Efficiency Comparison

This section delves into the computational complexity of the proposed D3PM in comparison with other methods. Specifically, we tabulated the number of learnable parameters (Params) for all approaches, as well as the average inference time required to process 100 samples from the MT-Wuhan dataset (time). The intuitive comparisons are presented in Table VI. Due to its dual-stream network design and the integration of multitimesteps and multiscale features, compared with the methods based on CNN and Transformer, D3PM exhibits a longer inference time and a larger parameter scale. In contrast to the DDPM-CD method that also uses DDPM, D3PM has a faster inference speed and fewer parameters. In conjunction with Fig. 6 and Table II, the D3PM model, though characterized by a greater computational demands, is also capable of achieving more precise predictions and enhanced generalization capabilities. This suggests that the D3PM is better suited for applications where stringent prediction accuracy is required, with additional computational resources leading to significant performance advantages. Considering the model's notable contribution to improving

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TABLE VI
NUMBER OF LEARNABLE PARAMETERS AND THE TIME REQUIRED FOR A SINGLE INFERENCE ACROSS DIFFERENT METHODS

Backbone	Method	Time(ms)	Params(M)
CNN	FC-EF(2018)	35.50	21.55
	FC-Siam-Diff(2018)	34.85	21.55
	FC-Siam-Conc(2018)	34.88	24.68
	SUNet(2021)	34.20	15.58
	SNUNet(2022)	39.35	12.03
	DA-MSCDNet(2022)	79.72	45.81
	ELGCNet(2024)	53.04	10.57
Transformer	ChangeFormer(2022)	55.57	41.02
	SwinSUnet(2022)	44.11	42.95
	ScratchFormer(2024)	92.38	36.91
	CMMAN(2024)	58.42	58.61
	MMCD(2024)	39.23	11.66
Diffusion model	DDPM-CD(2024)	259.23	434.94
	D3PM(Ours)	164.61	74.94

detection accuracy, its increased computational requirements are justified in many practical applications.

V. CONCLUSION

This article proposes a dual-stream network based on DDPM for remote sensing image MCD. The diffusion models are known for their ability to learn features during the denoising process, to leverage this characteristic, we designed an unconditional DDPM stream with the aim of achieving robust feature extraction; inspired by the success of diffusion models in image translation, we also designed an conditional DDPM stream aimed at improving comparability of multimodal data features. Furthermore, we devised a collaborative learning optimization strategy, which not only maintains the integrity of learning for both the denoising and CD tasks but also facilitates cross-task knowledge sharing. Benefiting from the dual-stream architecture, heterogeneous images can be transformed into comparable semantics in a common latent space during the denoising and translation processes, and these semantics are used as key features for predicting change maps. Experiments on public datasets show that D3PM outperforms several other methods in terms of accuracy. Ablation experiments validate the rationality and effectiveness of the D3PM design.

Despite the encouraging results, the D3PM framework still has room for further improvement and extension.

1) *Dependency on Labeled Data:* The D3PM framework relies on high-quality and diverse labeled data, which restricts its performance on large-scale unlabeled data. Moreover, insufficient coverage of various change types and modality differences in the training data also limits the model's generalization capability to new tasks.
2) *Computational Complexity:* Incorporating a dual-stream U-Net encoder and multilayer decoder results in high computational costs, potentially hindering deployment in resource-constrained environments.
3) *Extreme Modality Differences:* Although D3PM is designed to handle modality heterogeneity, it may still face challenges when there are extreme differences between imaging modalities.

Overall, the proposed method not only establishes semantic feature dependencies between multimodal data but also mitigates the impact of varying distributions in heterogeneous data on CD tasks. We will continue to explore the potential of diffusion models in the remote sensing multimodal domain in the future.

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